Martin Long Hoang Tran

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Software Engineer Intern

EDUCATION

Cal State University, Fullerton - Bachelor of Science, Computer Science **Expected in Dec 2027**

EXPERIENCE

Software Development Intern, OpenQQuantify - Vienna, VA May 2025 - Oct 2025

- Developed an AI-powered chatbot, improving FAQ accuracy by **50%** and reducing response time by **10 seconds**.
- Built a Python ETL pipeline processing **50K+ drug label records**, reducing manual data preparation by **15 hrs/week**.
- Designed and implemented RESTful APIs and a responsive frontend for real-time user interaction in an **Agile** development environment using **Git**-based workflows (feature branching, version control).

PROJECTS

RevenueTwin | AI Revenue Audit Platform | LPL Financial Hackathon (1st Place Track Winner)

- Engineered an **Amazon Bedrock** agent using **boto3** to orchestrate multi-turn tool calls without delegating fee math.
- Hardened agent against cross-case context bleed and hallucination via prompt guardrails and output normalization.
- Architected strict service boundaries isolating **Bedrock** reasoning from engine math, enforcing read-only tool access.
- Constructed **CLI smoke-test** harnesses and **unit tests** to validate agent schema compliance and tool dispatch.
- Github: https://github.com/martintran23/RevenueTwin

WeCalendar | Shared Calendar & Notes Platform | Full-Stack Web Application

- Designed **PostgreSQL** schema with **RLS** across 10 tables, workspace invite **RPCs**, and 9 SQL migrations.
- Engineered **Supabase** real-time synchronization across 7 tables for live multi-user calendar, tag, and notes updates.
- Built client-side conflict detection with recurrence grouping, replacing DB constraints with UI warnings.
- Implemented collaborative Notes with **TipTap** rich text, shared/private **RLS**, folders, and calendar event linking.
- GitHub: https://github.com/WeCalendar1/WeCalendar

WattTime | Solar Energy Optimization Platform | CloudHacks 2026 @ UCI

- Built a full-stack application that optimizes appliance usage using solar forecasts and location-based data.
- Designed an event-driven pipeline with **AWS API Gateway** and **Step Functions** for scalable optimization workflows.
- Reduced response time by **66% (45s → 15s)** by refining LLM prompts and improving inference efficiency.
- Implemented geocoding with **Amazon Location Service** and deployed via **AWS Amplify** with CI/CD.
- GitHub: https://github.com/martintran23/zotwatt

PROFESSIONAL DEVELOPMENT

System Engineer Participant, NASA L'SPACE Program - Tempe, AZ May 2025 - Aug 2025

- Researched technologies aligned with NASA's Technology Taxonomy, including radiation modeling, mitigation, and life support systems for long-duration missions.
- Analyzed systems engineering trade-offs for spacecraft safety, survivability, and mission-critical operations.
- Evaluated technical proposals using NASA review criteria and provided structured feedback.
- Designed and refined CAD models using **Siemens NX** to support system-level concepts.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL
Frameworks & Tools: Next.js, React, Node.js, FastAPI, Flask, Tailwind, Vitest, Postman, Git, Streamlit, boto3
Databases: MongoDB, PostgreSQL, Supabase
Cloud & DevOps: AWS (Bedrock, S3, API Gateway, Step Functions, Amplify), Vercel, REST APIs, CI/CD, Docker, Linux/Unix